

Rethinking.Logistics

Urban Logistic Hub

Verbitska Khrystyna



AD4
2026

Kurilla/Prokop studio

ARCHIP

ARCHITECTURAL
INSTITUTE
IN PRAGUE

Studio brief

Rethinking.Logistics explored how warehouses can become positive parts of the city instead of harmful ones.

Working with industry partners including Ramboll, Henning Larsen, and Panattoni, the studio addressed real-world challenges of logistics development, we focused on designing logistics buildings using data and measurable impacts like energy, carbon, and biodiversity.

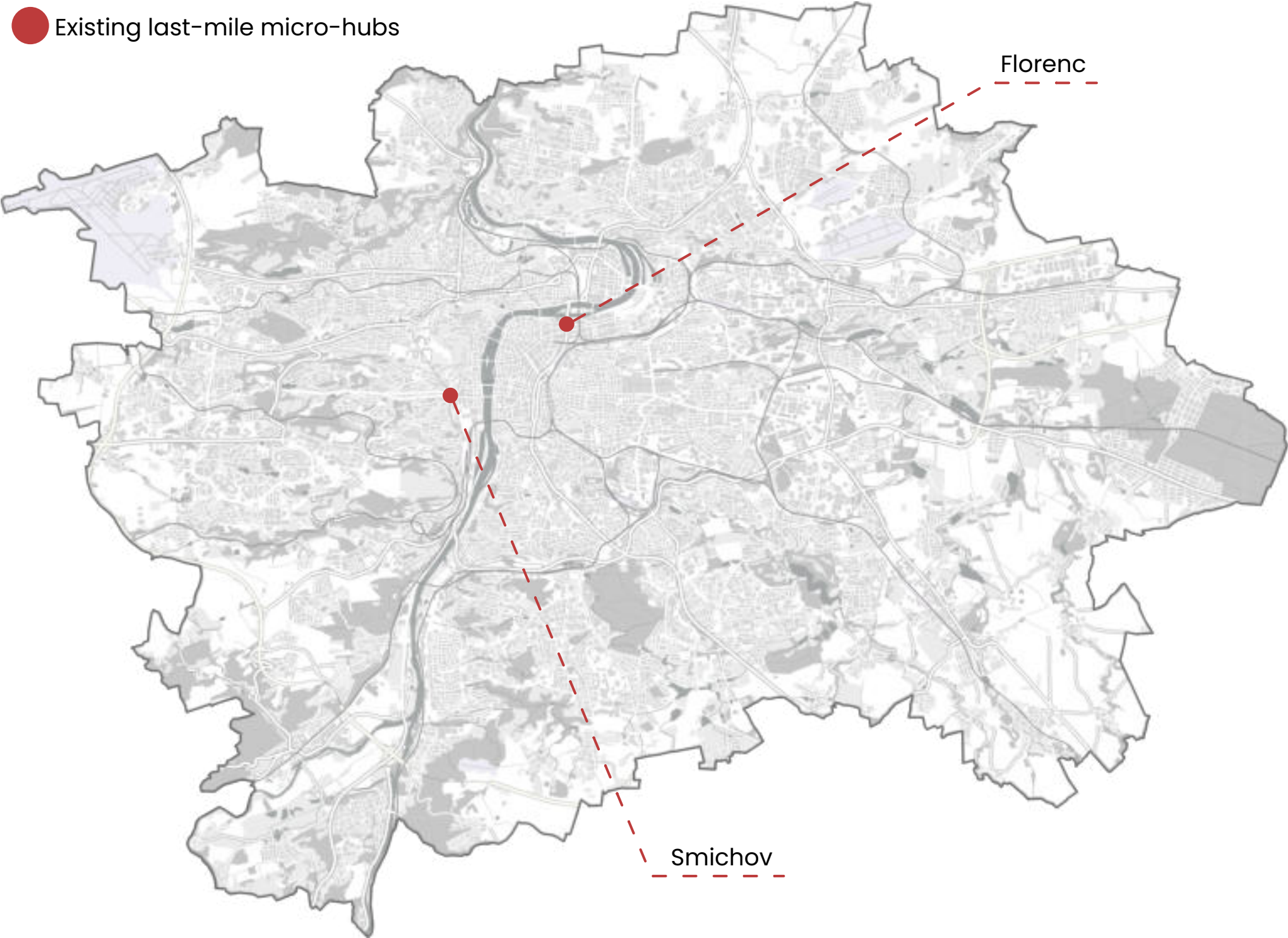
The studio combined collaboration with professionals and parametric tools to balance efficiency with environmental and social value.

Content

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Context

● Existing last-mile micro-hubs



Prague is one of the fastest-growing e-commerce and logistics markets in Central Europe, yet the city currently operates only a very limited number of urban micro-hubs dedicated to sustainable last-mile delivery.

Problems

Last-mile delivery is one of the most challenging and expensive parts of logistics. Mostly all deliveries go from warehouses outside the city, which causes problems such as:

Sustainability



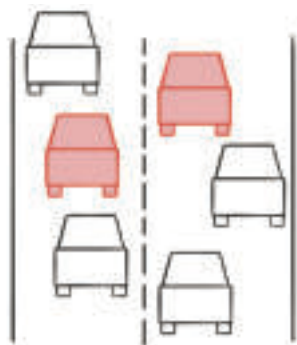
- Fossil fuel dependency;
- High CO₂ emissions;
- 30–50% of logistics-related pollution in dense cities;

Urban integration



- Large inactive industrial surfaces;
- Noise and visual pollution;
- Minimal community

Traffic



- Increased travel times;
- Increased road occupation during peak hours;
- 15–25% of urban traffic flows;

Accessibility

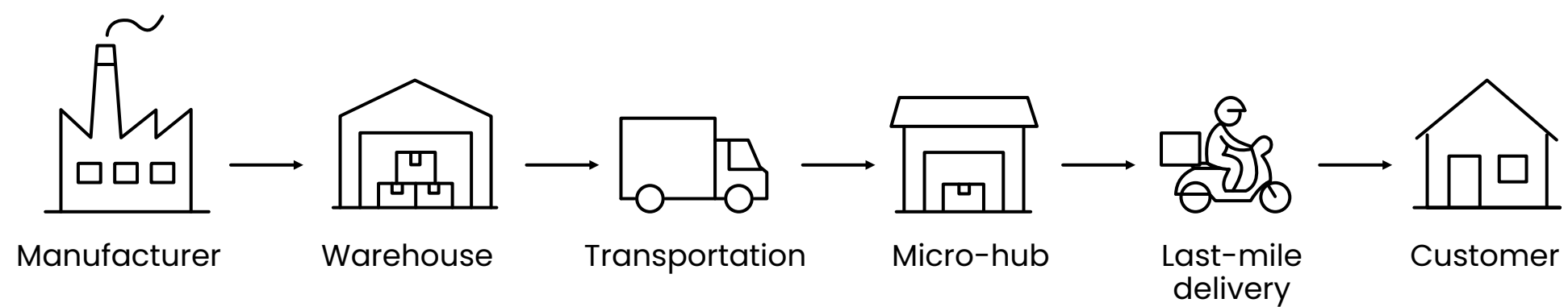


- Loading in pedestrian zones;
- Blocked sidewalks;
- Reduced comfort in public spaces;

Supply chain

The micro-hub system shortens delivery routes by introducing localized distribution points closer to customers.

Cargo bikes and electric mobility improve efficiency while reducing emissions and traffic pressure.



End-to-end cycle time
64% faster

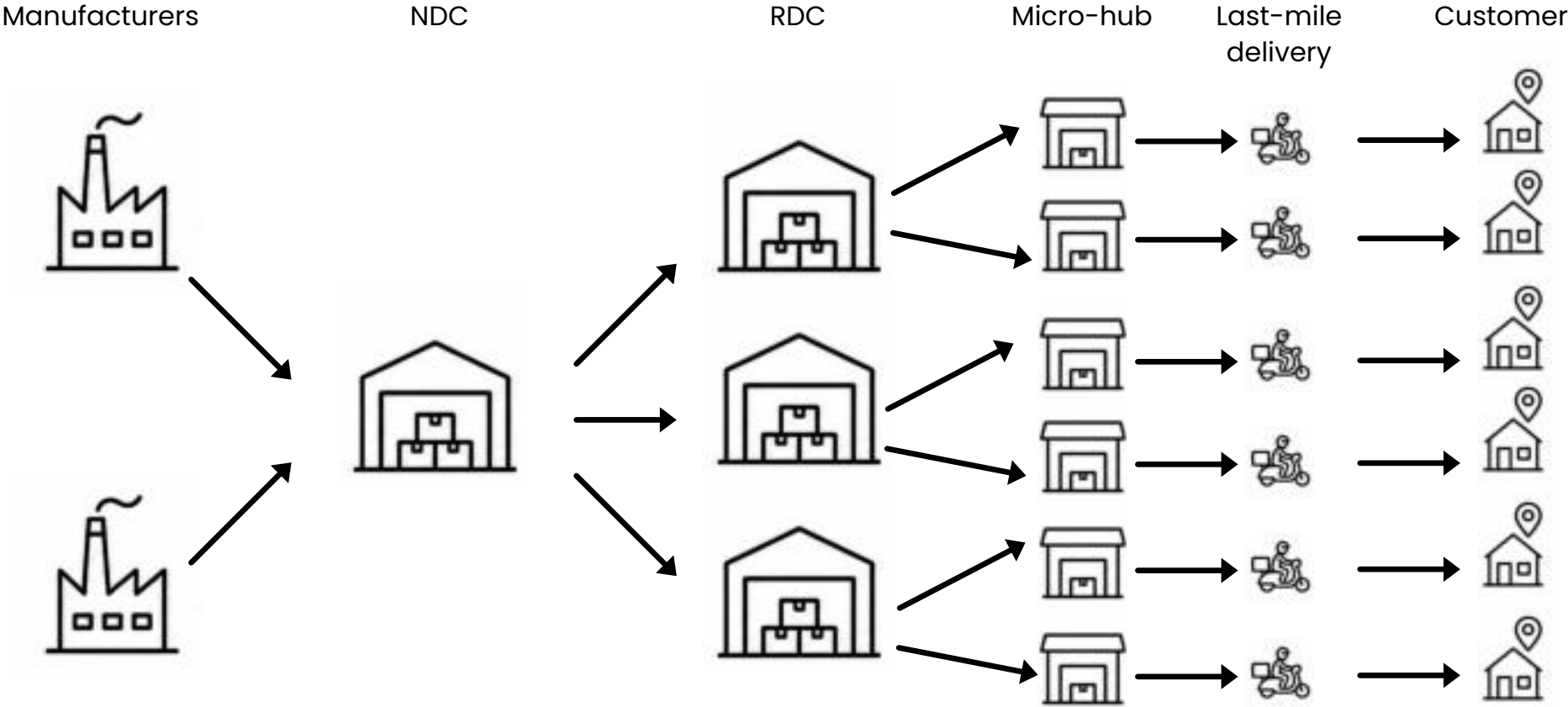
Shorter routes enable faster deliveries and better reliability.

Supply chain cost
34% lower

Reduced transport, optimized routes and lower handling costs.

CO₂ emissions
40% reduction

Clean mobility and localized distribution significantly cut emissions.





NDC

Receives goods from manufacturer and consolidates shipments



RDC

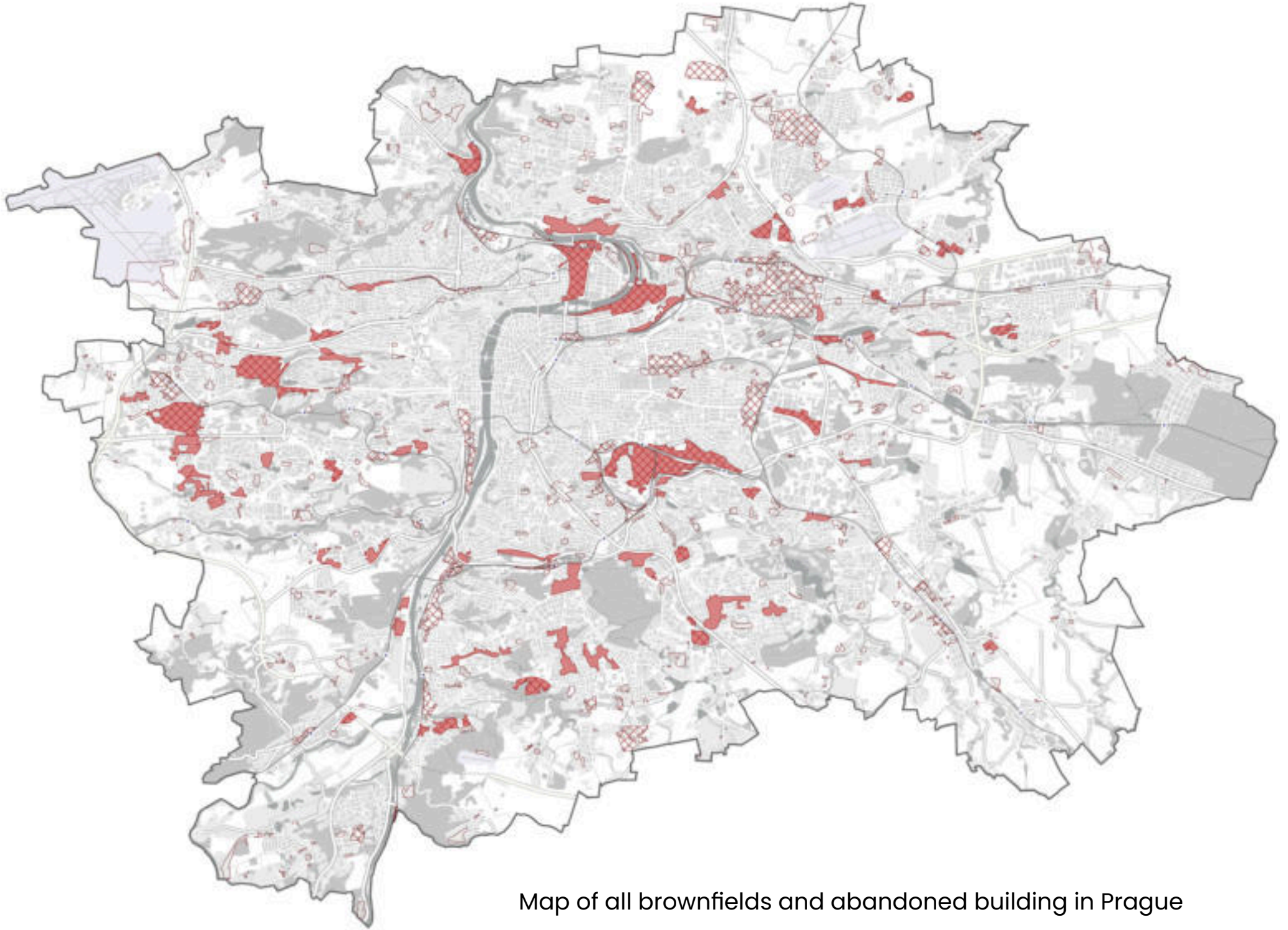
Distributes goods to micro-hubs based on local demand.



Micro-hub

Urban hub for storage, sorting and cargo-bike dispatch within a small radius.

Concept



Map of all brownfields and abandoned building in Prague

Instead of expanding logistics infrastructure into undeveloped land, the project reactivates existing urban space and transforms it into last-mile logistics hub integrated within the city fabric.

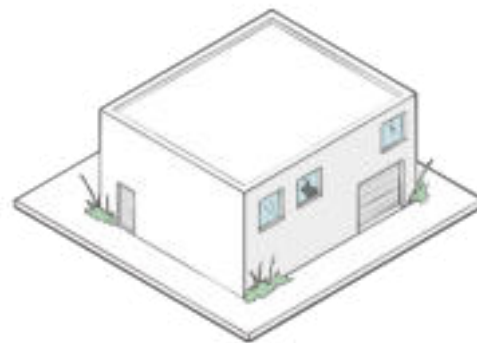
Concept

Last-mile micro-hub in Prague, that consolidates, sorts and distributes goods by EV- cars and cargo- bikes.

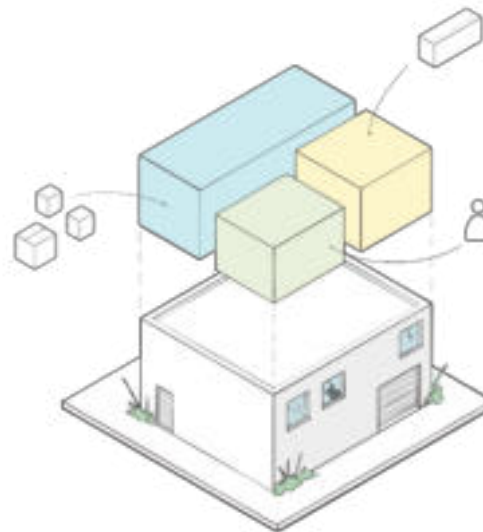
The project reuses existing urban structure, which categorized as abandoned building, to create an efficient, low-impact logistics model that supports local business, community space and a greener city,



Existing site



Abandoned building



Adaptive mixed-use



Integrated hub

Logistics Hub

Micro-hub center for parcel sorting, cargo-bike dispatch, and efficient low-emission last-mile delivery.

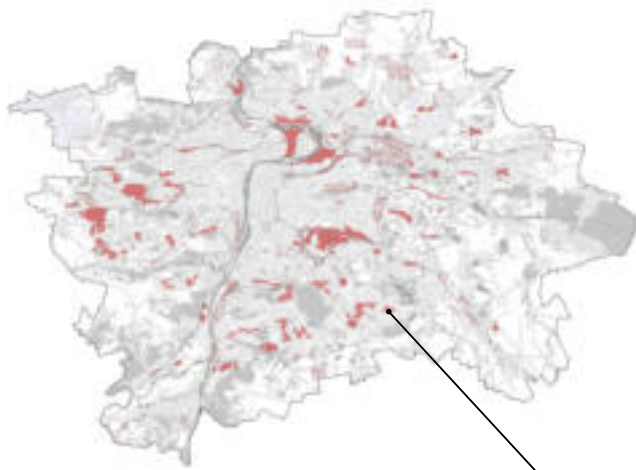
Showroom

A flexible showroom space combining product display, repair facilities, and community-oriented retail functions.

Community Garden

Community garden for public interaction that reconnects logistics infrastructure with urban life.

Location



Prague 11 - Haje

The project is located within a former industrial area characterized by abandoned infrastructure and underused urban land. The existing structure provides a strong concrete framework suitable for adaptive reuse, reducing the environmental impact associated with demolition and new construction.

Its strategic urban location enables efficient last-mile distribution while maintaining close connections to surrounding residential and commercial districts. The site benefits from existing transport infrastructure, accessibility, and proximity to dense urban activity, making it ideal for a decentralized logistics model.

Design principle

Proposal 1



The proposal investigates a more hybrid spatial model in which logistics, mobility, and community programs coexist within the same environment.

The overlap of functions creates a more transparent and socially integrated system

. At the same time, the project addresses the challenges of mixed-use coexistence through controlled circulation, spatial thresholds, and gradients of intensity that balance operational efficiency with public accessibility.

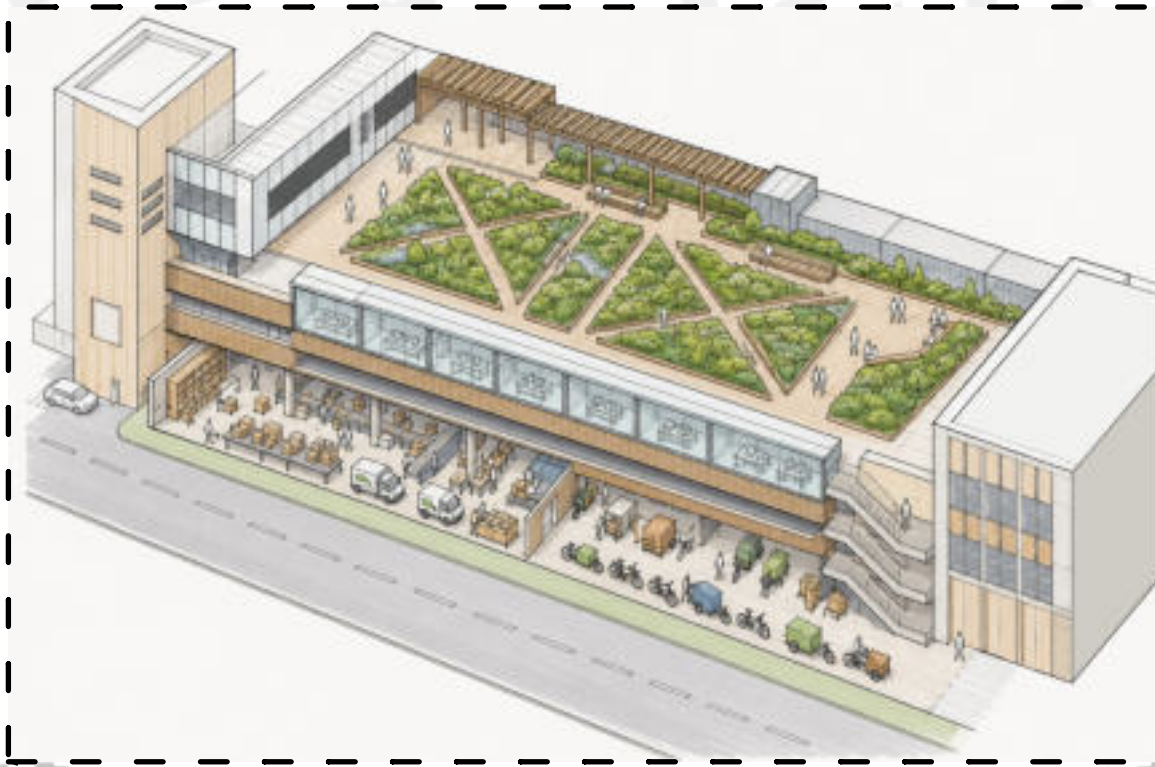
Proposal 2

Mixed-use functions separated by floors mainly to avoid operational conflicts while still keeping the building socially integrated into the neighborhood.

Ground / logistics layer for cargo bike dispatch, EV and bike servicing infrastructure, loading and unloading.

Public / community layer for showroom, light retail functions, community space

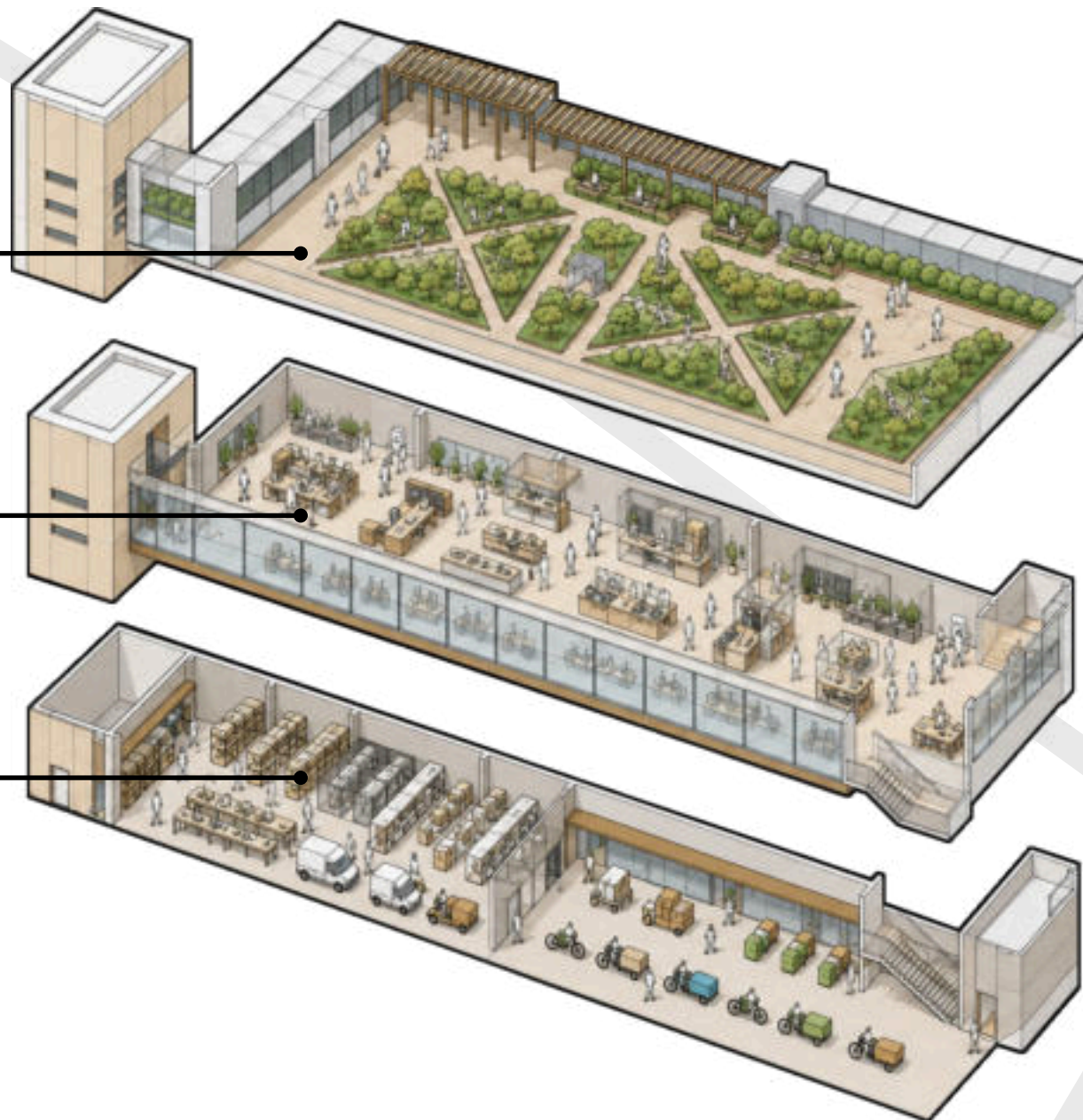




Open garden

Showroom

Last-mile delivery



For nature



For people



For the city



For business

Urban Biodiversity

The introduction of rooftop greenery and planted public spaces increases urban biodiversity, improves microclimate, and supports rainwater retention.



+35% Green Surface Area

Existing condition:
mostly sealed industrial surfaces

Proposed intervention:

approx. 210 m² of new green surfaces added to a 600 m² site

increase of approximately 35% vegetated area



Existing building area = 600 m²

New construction embodied carbon:
500 kg CO₂e/m²

Reuse scenario saves approx. 70% of new
construction carbon.

$600 \text{ m}^2 \times 500 \text{ kg CO}_2\text{e/m}^2 = 300,000 \text{ kg CO}_2\text{e}$

$300,000 \times 70\% = 210,000 \text{ kg CO}_2\text{e saved}$

Carbon Reduction

By reusing an abandoned building, the project reduces embodied carbon from demolition, new materials, and construction.

Community integration

The project transforms closed industrial infrastructure into an active public destination that supports social interaction, accessibility, and community engagement.



+210 m² of Accessible Community Space

Existing condition:

abandoned industrial building

0 m² publicly accessible space

Proposed intervention:

rooftop public garden → ~150 m²

showroom & interaction zone → ~60 m²

